Hybrid physics: SPH where the flow is violent, 2D shallow water where the river is long.

TECH STACK

Solvers
In-house C#: SWE-2D finite volume (HLL) · WCSPH
Adapters: Delft3D FM · DualSPHysics (GPU)

Geo
GeoTIFF DEMs (GLO-30 / SRTM) · NetTopologySuite
Priority-flood + D8 · OpenStreetMap · Earth Engine

Backend
.NET 8 · Clean Architecture · CQRS (MediatR)
Hangfire + Redis · Neon PostgreSQL · JWT auth

Frontend
React · MapLibre (3D terrain) · Recharts
SignalR live progress

Deploy
Docker Compose · Kubernetes-ready (stateless API)
Offline simulation core (cached terrain)

PIPELINE

1. Input
Pick a dam from the register, place a custom dam, or draw a landslide dam on satellite imagery

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2. Auto-preprocessing
DEM clip, river tracing, sill carving, area-elevation-volume curve, Manning's n

↓

3. Breach model
Froehlich 2008 / MacDonald-LM (dams), Costa & Schuster (landslide) + reservoir routing

↓ breach hydrograph

4a. SPH near field (WCSPH / DualSPHysics)
Breach + first 1-3 km, Froude-scaled 2D section
Violent flow, debris mode, impact force on barrage

handover →

4b. 2D far field (SWE-2D / Delft3D FM)
20-100 km downstream, shallow water equations
Cascade dams fail in-grid · Monte Carlo ensemble

↓

5. Post-process + comparison
Max depth, velocity, arrival time, hazard H1-H6 (depth x velocity), SPH vs SWE vs Ritter in overlap zone, flood-probability map

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6. Impact + HADR decision support
Population, bridges, roads, loss in Rs, golden time, helipads, sensor sites

↓

7. Export + Dashboard .shp, .kmz (animated), GeoJSON, OGC WMS, PDF sitrep, CAP 1.2 alert

validation ↑

GEE (near real-time)
Sentinel-1 flood map → CSI score
Sentinel-2 NDWI: new lake found → auto-simulate → CAP alert

METHODOLOGY

Breach
Froehlich (2008) breach parameters

$$Q_p = 0.607 \cdot V_w^{0.295} \cdot h_w^{1.24}$$

peak outflow cross-check, Froehlich (1995)
Costa & Schuster (1988) for landslide dams

Why hybrid?
SPH: violent flow, debris and impact force near the breach
SWE-2D / Delft3D: shallow water routes the flood 100 km downstream, in minutes

Validation ladder
1. **Ritter / Stoker** analytic — `L1 2.9% / 0.8%`
2. **Malpasset 1959** benchmark — `runner ready`
3. **Teesta 2023** vs Sentinel-1 — `CSI pending`